Bounded Schauder matrices are path-connected, and the unconditional orbit is not closed by basis type

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Lane 13, model GPT5.6

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Status. Candidate full solution, likely valid, subject to human review. The first theorem affirmatively answers Question 3.8 of the source. The second theorem gives a negative answer to Question 3.10, already when the starting matrix is the identity (and hence unconditional).

1 Source questions

The source is Yang Cao, Geng Tian, and Bingzhe Hou, *Schauder Bases and Operator Theory III: Schauder Spectrums*, arXiv:1204.4232. Fix an orthonormal basis (e_n) of an infinite-dimensional separable complex Hilbert space $\mathcal{H}$. The paper calls the matrix whose columns form a Schauder basis a *Schauder matrix*. In Remark 3.7 it restricts to those matrices that represent bounded operators and equips them with the operator-norm topology. We denote this set by $\mathcal{F}$.

Question 3.8 asks whether $\mathcal{F}$ is connected. For $F \in \mathcal{F}$, the source defines

$$\mathcal{O}_{\mathrm{gl}}(F) = \{XF : X \in \mathrm{GL}(\mathcal{H})\}$$

and lets $\mathcal{O}^{c}_{\mathrm{gl}}(F)$ consist of the Schauder matrices that are norm limits of elements of $\mathcal{O}_{\mathrm{gl}}(F)$. Question 3.10 asks whether conditionality or unconditionality must be preserved upon passing to $\mathcal{O}^{c}_{\mathrm{gl}}(F)$.

2 Proof intuition

The source already observes that every bounded Schauder operator T is injective with dense range. Therefore its polar decomposition $T = U|T|$ has a *unitary*, rather than merely partially isometric, polar factor U. Left multiplication by a unitary preserves the basis property. We can rotate U continuously to the identity, leaving the positive Schauder operator $|T|$. The straight line from $|T|$ to the identity is even simpler: except at its initial endpoint, every point on it is positive and bounded below, hence invertible. This joins every member of $\mathcal{F}$ to the identity.

The same polar decomposition shows more. For every $\varepsilon > 0$, $U(|T| + \varepsilon I)$ is invertible and is exactly ε away from T. Thus *every* bounded Schauder matrix lies in the closure of the orbit of the identity. Since bounded conditional Schauder matrices exist, the closure of this unconditional orbit does not preserve unconditionality.

satisfying the following properties:
1. for each $t \in I$, $\gamma(t)$ is a Schauder matrix;
2. $\gamma(t)$ represents a bounded operator;
3. The map is continuous in the variable $t \in I$ under the norm topology $L(\mathcal{H})$.
Here I is an interval(either open or closed). Denote by $\mathcal{F}$ the set of all Schauder matrices, we have the following question:

Question 3.8. Does $\mathcal{F}$ must be a connected set?

Given a Schauder matrix F, denote by $O_{gl}(F)$ the set consisting of all Schauder matrices equivalent to F. As well-known, invertible operators are connected(Problem 141, [10], p76), so we have

Theorem 3.9. *The set $O_{gl}(F)$ is always path-connected for each Schauder matrix F.*

Denote by $O^c_{gl}(F)$ the set of Schauder matrices F which is a Schauder matrix and there is a sequence $F_k \in O_{gl}(F)$ such that $||F_k - F|| \to 0$.

Question 3.10. If F is a conditional(unconditional) matrix, whether each matrix $F^{'} \in O^c_{gl}(F)$ must be also a conditional(unconditional) matrix or not?

Figure 1: Questions 3.8 and 3.10 on source PDF page 6.

3 Path-connectedness

Theorem 1. *The set $\mathcal{F}$ of bounded Schauder matrices on $\mathcal{H}$ is path-connected in the operator norm. More precisely, every $T \in \mathcal{F}$ can be joined to I by a norm-continuous path contained in $\mathcal{F}$.*

Proof. Let $T \in \mathcal{F}$ and write its polar decomposition as

$$T = UA, \qquad A = |T| = (T^*T)^{1/2}.$$

Because the columns (Te_n) form a Schauder basis, T is injective and has dense range. Hence $\ker A = \{0\}$, $\overline{\operatorname{ran} A} = \mathcal{H}$, and the polar factor U is unitary. Moreover,

$$Ae_n = U^*Te_n,$$

so (Ae_n) is a Schauder basis: it is the unitary image of (Te_n).

By the spectral theorem there is a bounded self-adjoint operator B such that $U = e^{iB}$. Define, for $0 \le t \le 1$,

$$\gamma_T(t) = \begin{cases} e^{i(1-2t)B}A, & 0 \le t \le \frac{1}{2}, \\ (2-2t)A + (2t-1)I, & \frac{1}{2} \le t \le 1. \end{cases}$$

The two formulas agree at $t = 1/2$, and functional calculus makes the path norm-continuous. Also $\gamma_T(0) = UA = T$ and $\gamma_T(1) = I$.

For $0 \le t \le 1/2$, the operator $e^{i(1-2t)B}$ is unitary, so the columns of $\gamma_T(t)$ are a unitary image of the Schauder basis (Ae_n). Consequently $\gamma_T(t) \in \mathcal{F}$.

For $1/2 < t \le 1$, put $s = 2t - 1 > 0$. Since $A \ge 0$,

$$(1-s)A + sI \ge sI.$$

Thus $\gamma_T(t)$ is positive and boundedly invertible, and its columns form a Riesz basis. At $t = 1/2$ the columns form the Schauder basis (Ae_n). Hence the entire path lies in $\mathcal{F}$, proving the theorem. $\square$

4 The orbit-closure question

Theorem 2. *For the identity Schauder matrix,*

$$\mathcal{O}^c_{\mathrm{gl}}(I) = \mathcal{F}.$$

In particular, Question 3.10 has a negative answer: an unconditional Schauder matrix can have a conditional Schauder matrix in the norm closure of its equivalence orbit.

Proof. The inclusion $\mathcal{O}^c_{\mathrm{gl}}(I) \subseteq \mathcal{F}$ is part of the definition of $\mathcal{O}^c_{\mathrm{gl}}(I)$. Conversely, take $T \in \mathcal{F}$ and again write $T = UA$ as above. For every $\varepsilon > 0$, set

$$T_\varepsilon = U(A + \varepsilon I).$$

The positive operator $A + \varepsilon I$ is boundedly invertible, as is T_ε. Therefore

$$T_\varepsilon \in \mathrm{GL}(\mathcal{H}) = \mathcal{O}_{\mathrm{gl}}(I).$$

Furthermore,

$$\|T_\varepsilon - T\| = \|U(A + \varepsilon I) - UA\| = \varepsilon.$$

It follows that $T \in \mathcal{O}^c_{\mathrm{gl}}(I)$ and hence $\mathcal{O}^c_{\mathrm{gl}}(I) = \mathcal{F}$.

For completeness, we now exhibit a bounded conditional member of $\mathcal{F}$. Let (u_n) be any conditional Schauder basis of $\mathcal{H}$, and choose positive numbers

$$\alpha_n = \frac{2^{-n}}{1 + \|u_n\|}.$$

The rescaled sequence $(\alpha_n u_n)$ is still a Schauder basis, with exactly the same partial-sum projections as (u_n), and is therefore conditional. Define S on the fixed orthonormal basis by

$$Se_n = \alpha_n u_n.$$

Since

$$\sum_n \|Se_n\|^2 \le \sum_n 4^{-n} < \infty,$$

S is Hilbert–Schmidt and in particular bounded. Thus $S \in \mathcal{F}$, and the first part of the theorem gives $S \in \mathcal{O}^c_{\mathrm{gl}}(I)$. But I is unconditional whereas S is conditional, which is the required counterexample. □

5 Verification and scope

The argument is internal to standard Hilbert-space operator theory. The key checks are:

1. dense range makes the polar factor unitary;
2. left unitary multiplication preserves all Schauder partial-sum projections up to unitary conjugacy;
3. $(1 - s)A + sI \ge sI$ for $A \ge 0$ and $s > 0$;
4. $U(A + \varepsilon I)$ is invertible and converges to UA in norm;
5. diagonal rescaling by nonzero scalars does not change the partial sums of a Schauder expansion.

The result concerns the bounded-operator class explicitly singled out in Remark 3.7 and the norm topology used there. It is stated for complex Hilbert space, matching the source's spectral setting. The counterexample settles the preservation question negatively; it does not assert that the closure of the orbit of every conditional basis contains an unconditional basis.

6 Novelty check

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:1204.4232, the exact title, "Schauder matrix", connectedness, and conditional orbit closure; no duplicate was found. Bounded web searches on 9 August 2026 used the exact wording and numbering of Questions 3.8 and 3.10, the paper title, and the authors. They found the source and adjacent papers on Schauder operators, but no later answer to either question.

The proof uses only the polar decomposition and a standard rescaling of a conditional basis, so it may be folklore or an unnoticed immediate consequence of known facts. The novelty assessment is therefore *plausibly new as an explicit answer to these questions, moderate confidence.*

7 Human-review recommendation

Recommend priority review. The highest-value checks are that the source's set $\mathcal{F}$ indeed fixes the same orthonormal coordinates along a path, and that its notation $\mathcal{O}^{c}_{\mathrm{gl}}(F)$ means relative norm closure among Schauder matrices as transcribed above. Under those definitions, both arguments are short and exact.

References

[1] Y. Cao, G. Tian, and B. Hou, *Schauder Bases and Operator Theory III: Schauder Spectrums*, arXiv:1204.4232 (2012).

[2] I. Singer, *Bases in Banach Spaces I*, Springer, 1970.

[3] A. M. Olevskii, *On operators generating conditional bases in a Hilbert space*, Math. Notes **12** (1972), 476–482.